

IPSEC REGULATORY AUDIT & CRYPTOGRAPHIC HEALTH

IPsec VPN Security Assessment & Framework Audit

Comprehensive evaluation against NIST SP 800-77 Rev. 1, RFC 8221, and NSA CNSA 2.0 cryptographic mandates

77
GRADE B+

CONDITIONAL COMPLIANCE · ATTENTION REQUIRED

Section 1: Scope & Regulatory Framework

Definition of audit benchmarks (NIST SP 800-77 Rev. 1, RFC 8221, NSA CNSA 2.0)

AUDIT SCOPE & TARGET GATEWAYS		ASSESSMENT TIMEFRAME	
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REGULATORY BENCHMARK	TITLE & JURISDICTION	MANDATED CRYPTOGRAPHIC REQUIREMENTS	AUDIT COMPLIANCE VERDICT

Section 2: Cryptographic Health Audit

Granular evaluation of symmetric ciphers (AES-GCM vs. CBC/3DES) and hashing integrity (SHA-256 vs. SHA-1/MD5)

SYMMETRIC ENCRYPTION CIPHER AES-256-GCM (AEAD) Operating Mode: Galois/Counter Mode (AEAD) Integrity Tag: 128-bit ICV (Authentication Tag) Key Length: 256-bit Symmetric Key AEAD provides built-in authentication, completely neutralizing padding oracle vulnerabilities. APPROVED	HASHING & MESSAGE INTEGRITY SHA-256 / HMAC-SHA2-256 Digest Size: 256-bit Digest Collision Resistance: Verified (FIPS 180-4 Compliant) Deprecated Primitives: MD5 / SHA-1 Not Detected Collision-resistant hashing adherence verified with zero deprecated digest algorithms detected. APPROVED
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KEY MANAGEMENT, PROTOCOL & STATE INTEGRITY

Key Exchange & Protocol Integrity

Diffie-Hellman modulus size verification, post-quantum posture, sequence rollover safeguards, and anti-replay integrity

Section 3: Key Management & Exchange Audit

Verification of Diffie-Hellman modulus size, post-quantum posture, and PFS rekey enforcement

DIFFIE-HELLMAN MODULUS STRENGTH APPROVED ECP-256 (DH Group 19) 256-bit Elliptic Curve (~3072-bit RSA Equivalent) Meets NIST SP 800-57 discrete logarithm strength recommendations for enterprise confidentiality.	PERFECT FORWARD SECRECY (PFS) REKEY CRITICAL NON-COMPLIANCE Disabled on Child SA IKE SA: Active (ECP-256) Penalty: -11 pts Child SA rekeying without ephemeral DH exchange allows retroactive bulk decryption if long-term credentials leak.	POST-QUANTUM CRYPTOGRAPHIC POSTURE TRANSITION REQUIRED Classical ECC (Pre-Quantum) Target: Hybrid ML-KEM-768 (Kyber) Advisable to adopt hybrid post-quantum key encapsulation per NSA CNSA 2.0 roadmap (2026-2030).	SA ROTATION LIFETIMES & LIMITS ALIGNED 3600 seconds (1 Hour) 1,000 MB (1 GB Transfer Threshold) Strict rekey intervals constrain cryptanalytic exposure window under high packet volumes.
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Section 4: Protocol & State Integrity

Anti-replay window sizing, sequence rollover safeguards, and IKE mode security (e.g., IKEv1 Aggressive Mode detection)

ANTI-REPLAY WINDOW SIZING ACTIVE & VERIFIED 64-Bit Sliding Window (RFC 4303) Kernel state maintains strict sequence bitmap. Monitored session accepted 0 duplicate or injected frames.	SEQUENCE ROLLOVER SAFEGUARDS HARDENED 64-Bit Extended Sequence Numbers (ESN) Protects against 2^32 sequence cycling attacks under 10Gbps+ multi-core line-rate traffic. Safe for >500 years.	IKE MODE SECURITY & DOWNGRADE PROTECTED IKEv2 (Main Mode Equivalent) IKEv1 Aggressive Mode NOT detected. Unencrypted identity hashing is prevented, neutralizing offline dictionary cracking.
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SIDE-CHANNEL AUDIT & COMPREHENSIVE VULNERABILITY REGISTER

Side-Channel Audit & Threat Matrix

Traffic Flow Confidentiality (TFC) leakage, itemized CVSS vulnerability register, and post-remediation posture

Section 5: Side-Channel & Metadata Vulnerability Audit

Leakage score based on packet shape predictability, burstiness, and missing Traffic Flow Confidentiality (TFC) padding

SIDE-CHANNEL LEAKAGE SCORE

74

MODERATE RISK

PACKET SHAPE PREDICTABILITY

HIGH PREDICTABILITY

Length clustering around 1420B and 180B leaks media application type.

BURSTINESS & IAT TIMING

MODERATE LEAKAGE

Regular 20ms inter-arrival intervals reveal active media codec timing.

TFC PADDING AUDIT

MISSING / DISABLED (-3 pts)

Activate random TFC padding (RFC 4303 §2.7) to obscure packet lengths.

Section 6: Comprehensive Threat & Vulnerability Matrix

Detailed register of discovered vulnerabilities, CVSS base/environmental scores, and audit remediation mandates

VULN ID	VULNERABILITY & THREAT VECTOR	CVSS v3.1 BASE / ENV	SEVERITY	REMEDIATION MANDATE
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PROJECTED POST-REMEDIATION POSTURE

CURRENT SCORE77 / 100

PROJECTED GAIN+19 pts

PROJECTED SCORE96 / 100

PROJECTED GRADEGRADE A+

Executing the top two priority directives (enforcing PFS Diffie-Hellman Group 19 on Child SAs and migrating to X.509 certificates) eliminates 16 penalty points, promoting the deployment into a fully hardened state compliant with NIST SP 800-77 Rev 1.